

1. The raw audio waveform consists of a very long list of amplitude values at various points of time. There is nothing that can tell us about the presence of particular frequencies in it. A neural network architecture such as CNN would be unable to extract any meaningful information from it. The spectrogram provides the temporal variation of frequencies, whereas the MFCC reduces it to an array of 13 numbers known as coefficients which represent the characteristic acoustics of the sound. These coefficients represent the acoustic pattern in a human-like manner and are more discriminative of sounds.
2. The full integer quantization requires both the weights and activations of the model to be represented using 8-bit integers instead of the standard 32-bit floating numbers. In order to achieve this with accuracy, the quantization process requires knowledge about the expected range of values for each activation that is going to be passed on during its operation. This is done using the representative data set which consists of a small portion of real input data.
3. The function `audio_processor.get_data()` fetches the dataset audio samples and performs pre-processing on them. This involves fetching the raw audio files in .wav format, resampling them to the proper sampling rate, ensuring that they are of equal length by adding silences or trimming off silences, calculating the MFCC values of the audio and finally, returns the feature arrays along with their respective labels.
4. The variable `g_model` is a C-byte array that holds the full TFLite model in hexadecimal form. This is the quantized neural network saved in the Arduino board's program memory. The second variable, `g_model_len`, stores the length of the C-byte array as an integer. Both variables must be copied accurately into `micro_features_model.cpp` file since they are used by TensorFlow Lite Micro to import and run the model on the Arduino board.

5.

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Heard unknown (244) @19520ms
Heard yes (209) @20416ms
Heard yes (204) @26976ms
Heard unknown (216) @28768ms
Heard unknown (254) @30304ms
Heard unknown (228) @31824ms
Heard unknown (248) @33360ms
Heard yes (214) @34512ms
Heard yes (202) @43680ms
```